BIRLA INSTITUTE OF TECHNOLOGY

B.Sc. (Artificial Intelligence & Data Science)

CURRICULUM BOOK

BASED ON NATIONAL EDUCATION POLICY 2020 (NEP-2020)

(Effective from Academic Session: 2025-26)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Institute Vision

To become a Globally Recognized Academic Institution in consonance with the social, economic and ecological environment, striving continuously for excellence in education, research and technological service to the National needs.

Institute Mission

- To educate students at Undergraduate, Postgraduate, Doctoral, and Post-Doctoral levels to perform challenging engineering and managerial jobs in industry.
- To provide excellent research and development facilities to take up Ph.D. programmes and research projects.
- To develop effective teaching learning skills and state of art research potential of the faculty.
- To build national capabilities in technology, education and research in emerging areas.
- To provide excellent technological services to satisfy the requirements of the industry and overall academic needs of society.

Department Vision

The department strives to be recognized globally for outstanding education and research, leading to excellent professionals and innovators in the field of Computer Science and Engineering, who can positively contribute to the society.

Department Mission

- To impart quality education and equip the students with strong foundation that could make them capable of handling challenges of the new century.
- To maintain state of the art research facilities and facilitate interaction with world's leading universities, industries and research organization for constant improvement in the quality of education and research.

Programme Educational Objectives (PEO)

- PEO 1: The program will produce graduates who will be competent professionals in IT industry, academics, government or entrepreneurs.
- PEO 2: Graduates will exhibit professional ethics, critical thinking, problem solving and effective communication skills to work collaboratively in a team-based environment.
- PEO 3: The graduates will possess leadership qualities and will be capable of attaining higher positions in their professional career.
- PEO 4: Graduates will be able to adapt to the fast-changing world of technology and will become effective professionals to address the technical, social and business challenges.
- PEO 5: Graduates will recognize the importance of interdisciplinary learning, engage in lifelong learning and professional development.

Programme Outcomes (PO)

- Discipline knowledge: Demonstrate the comprehensive knowledge of mathematics, computing fundamentals and domain concepts to enhance their professional skills.
- Problem analysis: Apply to identify, formulate and analyze solutions to various computing problems using the fundamental principles of computing.

- Design/development of solutions: Ability to design, develop and implement computer-based solutions to real world problems using appropriate tools and techniques.
- Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern IT tools including prediction and modeling to challenging problems.
- The graduates and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the IT professionals.
- Environment and sustainability: Understand the impact of the professional computer-based solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- Ethics: Apply ethical principles to maintain the integrity in a working environment in sustainable societal development through objective, unbiased and truthful actions.
- Individual and team work: Ability to work effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- Communication: Express thoughts and ideas effectively in understanding computing activities by writing effective reports, making effective presentations, constructing documentation and presenting complex information in a concise manner.
- Project management: Learn to build a project from pre-implementation to completion within constraints in multidisciplinary environments.
- Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSO)

- PSO 1: The ability to analyze, design, code, and test application-specific or complex problems in artificial intelligence, machine learning, big data analytics, cloud computing, and cybersecurity by applying knowledge of basic sciences, mathematics, and computing fundamentals.
- PSO 2: The ability to adapt to rapid changes in tools and technology with an understanding of societal and ecological issues relevant to professional practice through lifelong learning.
- PSO 3: Excellent adaptability to function in a multidisciplinary work environment, strong interpersonal skills as a leader in a team, and appreciation of professional ethics and societal responsibilities.

SEMESTER 1 - FIRST MONSOON

Total Credits: 21

Course Code	Course Name	L	T	P	Credits
SC25101	Programming with C (DSC)	3	0	0	3
SC25105	Operating Systems (DSC)	3	0	0	3
SC25107	Fundamentals of Computing (DSC)	2	0	0	2
SC25109	Fundamentals of Mathematics (MDC)	3	0	0	3
VAC	VAC Elective Course - I				2
VAC	VAC Elective Course - II				2
SC25102	C Programming Lab (DSC)	0	0	3	1.5
HS24131	Communication Skills - I (AEC)	0	0	3	1.5
SBC	SEC-SB Elective I	0	1	4	3

SEMESTER 2 - SECOND SPRING

Total Credits: 19.5

Course Code	Course Name	L	T	P	Credits
SC25113	Introduction to Data Structures (DSC)	3	1	0	4
SC25115	Python Programming (DSC)	3	0	0	3
SC25117	Mathematics for Computing (MDC)	3	0	0	3
VAC	VAC Elective Course - III				2

SC25114	Data Structures Lab (DSC)	0	0	3	1.5
SC25116	Python Programming Lab (DSC)	0	0	3	1.5
HS24133	Communication Skills - II (AEC)	0	0	3	1.5
SBC	SEC-SB Elective II	0	1	4	3

Total Credits after I year: 40.5

Vocational Courses for Exit after 1st Year

Category	Course Name	Code	L	T	C
Vocational Course I	Event Management	MN25120	3	0	3
Vocational Course II	Fundamentals of Multimedia	CA25133	3	0	3

SEMESTER 3 - THIRD MONSOON

Total Credits: 20

Course Code	Course Name	L	T	P	Credits
SC25201	Data Analytics and Visualization (DSC)	3	0	0	3
SC25203	Database Design Concepts (DSC)	3	0	0	3
SC25205	Mathematics for Artificial Intelligence (MDC)	3	0	0	3
MN25106	Principles of Management (MDC)	3	0	0	3
MN25109	Public Speaking and Creative Writing (AEC)	1	0	2	2
SBC	SEC-SB Elective III	2	0	2	3
SC25202	Data Analytics Lab (DSC)	0	0	3	1.5
SC25204	Database Design Lab (DSC)	0	0	3	1.5

SEMESTER 4 - FOURTH SPRING

Total Credits: 20

Course Code	Course Name	L	T	P	Credits
SC25211	Artificial Intelligence and Applications (DSC)	3	0	0	3
SC25213	Data Mining and Applications (DSC)	3	0	0	3
SC25215	Fundamentals of Computer Algorithms (DSC)	3	1	0	4
DSE	DSE I (Domain Specific Elective)	3	0	0	3
MN25201	Personality Development (AEC)	3	0	0	3
SC25214	Data Mining Lab (DSC)	0	0	3	1.5
SC25216	Computer Algorithms Lab (DSC)	0	0	3	1.5

Total Credits after II year: 80.5

Vocational Courses for Exit after 2nd Year

Category	Course Name	Code	L	T	C
Vocational Course III	Basics of Financial Markets and Equity Research	MN25214	3	0	3
Vocational Course IV	Business Intelligence	CA25225	3	0	3

SEMESTER 5 - FIFTH/THIRD MONSOON

Total Credits: 19.5

Course Code	Course Name	L	T	P	Credits
SC25301	Basics of Machine Learning (DSC)	3	0	0	3
SC25303	Software Engineering Lab (DSC)	0	0	3	1.5
SC25305	Deep Learning (DSC)	3	0	0	3
DSE	DSE II (Domain Specific Elective)	3	0	0	3
SC25302	Machine Learning Lab (DSC)	0	0	3	1.5
SC25304	Deep Learning Lab (DSC)	0	0	3	1.5
SC25308	Minor Project-1 / MOOC	0	0	0	3

SEMESTER 6 - SIXTH/FOURTH SPRING

Total Credits: 19.5

Course Code	Course Name	L	T	P	Credits
SC25315	AI and Generative Applications (DSC)	3	0	0	3
SC25319	Big Data Analytics (DSC)	3	0	0	3
SC25321	Cyber Security (DSC)	3	0	0	3
DSE	DSE III (Domain Specific Elective)	3	0	0	3
SC25316	AI Applications Lab (DSC)	0	0	3	1.5
SC25320	Big Data Analytics Lab (DSC)	0	0	3	1.5
SC25322	Project / Capstone	0	0	0	6

Total Credits after III year (Final): 119.5

Skill Enhancement Courses - Skill Based (SEC-SB)

Code	Subject	L	T	P	Credits
CA25110	Office Automation Tools	0	1	4	3
CA25112	Linux Administration	0	1	4	3
CA25130	Programming with MatLab	0	1	4	3
CA25132	Introduction to Latex	0	1	4	3
CA25209	Statistics with R	2	0	2	3
CA25215	Computer Oriented Numerical Methods	2	0	2	3

Value Added Courses (VAC)

Code	Subject	L	T	P	Credits
MN25102	Human Values and Professional Ethics	2	0	0	2
MN25103	Yoga	1	0	2	2
MN25104	Physical Education	1	0	2	2
CA25109	Environmental Science	2	0	0	2
MN25111	Digital Empowerment	2	0	0	2
MN25112	Emotional Intelligence	2	0	0	2
SC25131	Design Thinking	2	0	0	2

Multidisciplinary Courses (MDC)

Code	Subject	Credits
SC25109	Fundamentals of Mathematics	3
SC25117	Mathematics for Computing	3
SC25205	Mathematics for Artificial Intelligence	3
MN25106	Principles of Management	3